TAL KLEIN

Data Scientist | Machine Learning & Computer Vision

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SUMMARY

Data Scientist with hands-on experience in machine learning, NLP, causal inference, computer vision, and large-scale data pipelines. Looking to deepen professional expertise, take on more complex modeling challenges, and contribute to data-driven products at scale.

PROFESSIONAL EXPERIENCE

Data Scientist & ML Researcher
Socially Embedded Lab, Ben-Gurion University
Aug 2024 – Oct 2026

- Independently built an NLP pipeline turning 4,000+ Hebrew court verdicts into 30+ structured features with GPT-5-mini (0.92 F1), choosing LLM prompting over RAG and a fine-tuned two-stage model for accuracy
- Found Arab defendants were 15-20 percentage points more likely to be incarcerated for drug offenses (1,459 drug and 823 weapons cases), using propensity score matching in R
- Validated results with statistical analysis: sensitivity tests (Rosenbaum bounds, E-values), multiple imputation (MICE), and a check for differential LLM extraction error by defendant ethnicity

PROJECTS

Full-Text Study of 2.5M arXiv Papers

- Processed 2,553,787 PDFs (terabytes to tens of GB) and tested 26 content measures with interrupted time series: priority claims fell 28.3%, released-resource claims rose 58.9% (GitHub)

Wikipedia Search Engine

- Deployed a hybrid search engine (BM25, TF-IDF, PageRank) on GCP over 6.4M documents, cutting query latency 8x (9.8s to 1.24s) at Precision@10 of 0.63 (GitHub)

Computer Vision

- Multimodal pipelines: next-frame video prediction with BLIP-2, YOLOv5 and GPT-4 (GitHub) and extreme-weather classification with CLIP and hand-crafted features (GitHub)

LoRA/QLoRA Fine-Tuning of Gemma-3

- Ran 15+ ablations and built an SFT and retrieval-grounded QA pipeline, raising closed-book exact-match from 0.009 to 0.611 (GitHub)

EDUCATION

M.Sc. Software & Information Systems Engineering
Ben-Gurion University | Aug 2024 – Sep 2026 | GPA 92

B.Sc. Data Engineering
Ben-Gurion University | Oct 2021 – Aug 2025 | GPA 86

TECHNICAL SKILLS

Programming: Python, Java, SQL, R

ML & AI: Machine Learning, Deep Learning, NLP, Computer Vision, LLM Fine-Tuning, PyTorch, Hugging Face

Statistics: Causal Inference, Statistical Modeling, Time Series Analysis, A/B Testing

Data & Tools: Data Pipelines, ETL, GCP, Docker, Git

MILITARY & VOLUNTEERING

Communications & Operations NCO — IDF
Sep 2017 – May 2020

- Led a team of 8 soldiers running data and network operations
- Held personal responsibility for system upgrades and communications readiness, keeping operations running in a high-pressure environment

Volunteer Puppy Raiser (Foster Family Program)
Israel Guide Dog Center | Jan 2025 – Jun 2026

LANGUAGES

Hebrew: Native
English: Fluent